William Cook

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Education

Liberty University Lynchburg, VA
B.S. Computer Science: Software Engineering & Technology Entrepreneurship *Expected May 2028*
GPA: 3.78/4.0 | **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Database Design, Information Security, Network Architecture, Advanced C++

Technical Skills

Languages: C++, Python, JavaScript, TypeScript, Rust, SQL
Technologies: React, React Native, Node.js, Express, FastAPI, Tauri, PostgreSQL, MongoDB, MQTT, Docker, AWS EC2, Linux, Nginx, Git

Experience

Cole Design & Development LLC Dec. 2025 – Present
Software Engineering Intern *Hudson, OH*

- **Designed, built, and deployed** ColeCloud, a production IoT device-management platform using Node.js, React, PostgreSQL, MQTT, Docker, and AWS EC2; served as the **sole developer** and launched the initial system in **three weeks**.
- **Architected** a configuration-driven backend using JSON and PostgreSQL JSONB that supports four device types through shared application logic, enabling new device types to be added primarily through configuration.
- **Implemented** user audit, application error, and device logging to improve debugging and operational **observability**; validated deployments on AWS EC2 and maintain a live product demo used with prospective clients.
- **Secured** MQTT communication with per-device X.509 certificates and mutual TLS, limiting compromised credentials to individual devices; remediated an anonymous internet-facing WebSocket endpoint and developed Rust/Tauri commands for direct hardware configuration and control.

Missionize June 2025 – Sept. 2025
Full Stack Developer *Remote*

- **Developed** React Native interfaces for habit tracking, journaling, and user dashboards and built REST API endpoints using Python, FastAPI, and MongoDB.
- **Collaborated** on a six-person startup team using Git feature branches, pull requests, code review, and iterative development to design, implement, and deliver application features.

Projects

Self-Hosted Server Infrastructure | *Linux, Python, Nginx, systemd, Tailscale* Ongoing

- **Operate five self-hosted services** on a Raspberry Pi 5 Linux server with 1TB NVMe storage, including three Minecraft servers, a Discord bot, and a custom Magic: The Gathering application.
- **Developed** asynchronous Python services for log processing and event forwarding and configured Nginx, HTTPS, SSH-key authentication, Tailscale, systemd service management, networking, and backups.

Leadership & Awards

Competitive Programming Club, Officer | Liberty University Aug. 2024 – Present

- Teach C++ and Python to 20+ members through weekly data structures, algorithms, and contest-preparation sessions; solved **150+** algorithmic problems on Kattis.

Japanese Language Club, President | Liberty University Aug. 2026 – Present

- Lead a 25+ member organization and developed lujapanese.com on self-managed Linux infrastructure using Nginx, HTTPS, and Git.

ICPC Mid-Atlantic Regional — T-20th / 111 Division 1 Teams 2025
ICPC Mid-Atlantic Regional — 6th / 61 Division 2 Teams 2024